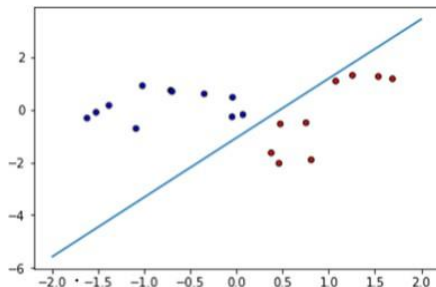


Linear Classifier

Classification Algorithms

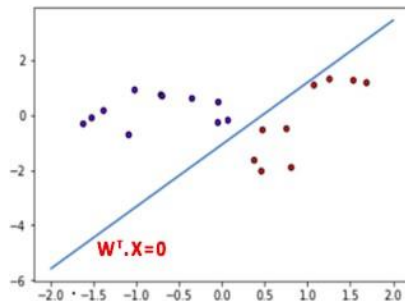
What is Linear?

- ▶ Recall the abstraction:
 - ▶ $y = f(w, x)$
- ▶ If f is linear, we have a Linear Classifier
- ▶ $f = \sum w_i x_i = W^T \cdot X$



Linear Classifier

- ▶ If $W^T \cdot X \geq 0$ then Class 1
- ▶ If $W^T \cdot X < 0$ then Class 0



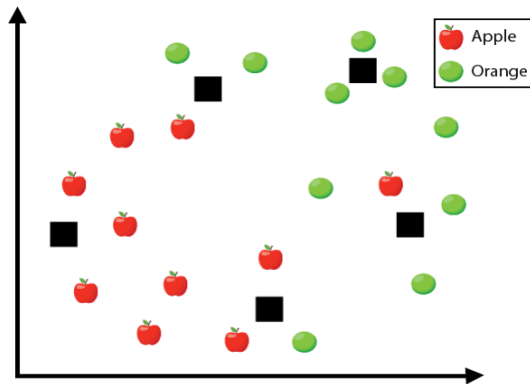
Linear Classifier

Data:

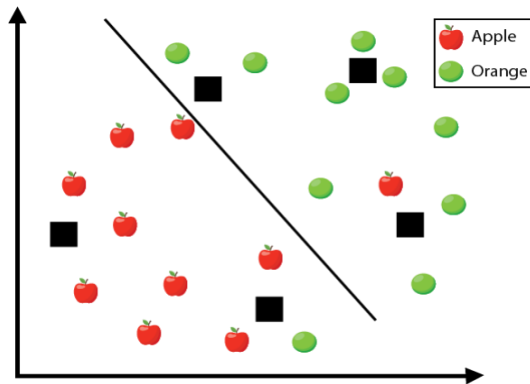
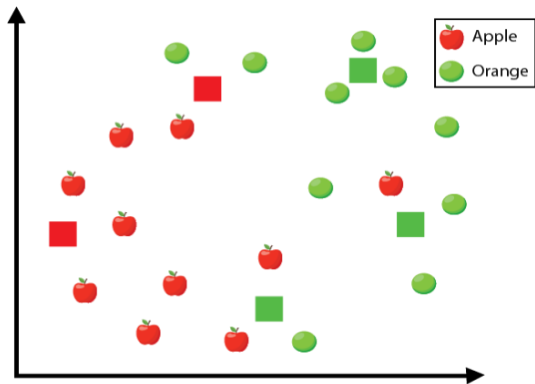
- ▶ Apples (red)
- ▶ Oranges (green)
- ▶ Test: Unknown (black)

Goal

- ▶ Find a line that separates



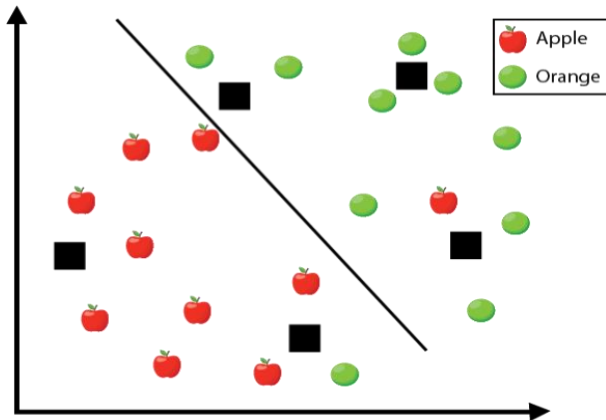
One Solution: $x_1 + 2x_2 + 2 = 0$



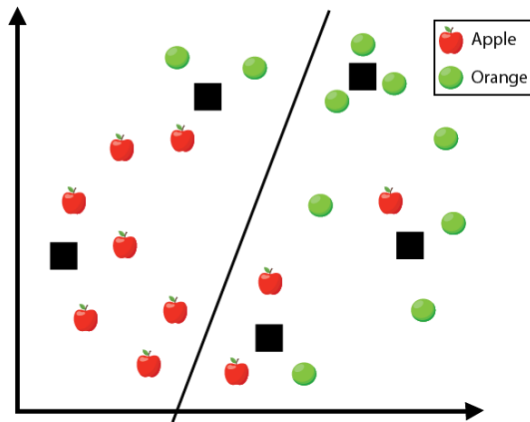
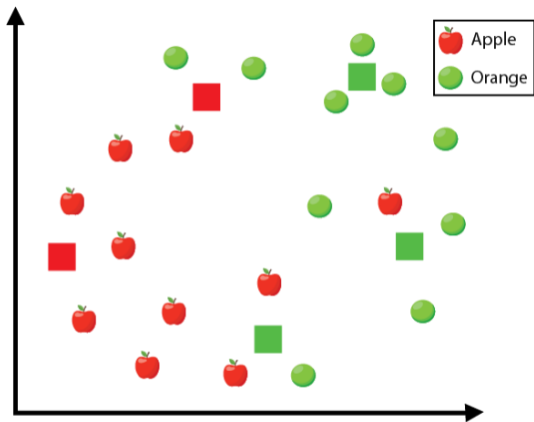
What is the Error/Accuracy?

On the test data:
40% error

On the training data:
10% error



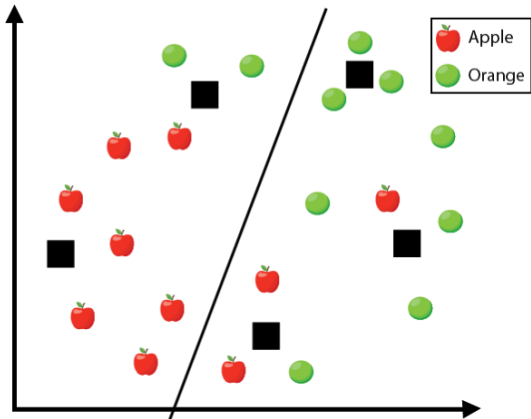
Best Solution: $2x_1 - 4x_2 - 5 = 0$



What is the Error/Accuracy?

On the test data:
0% error

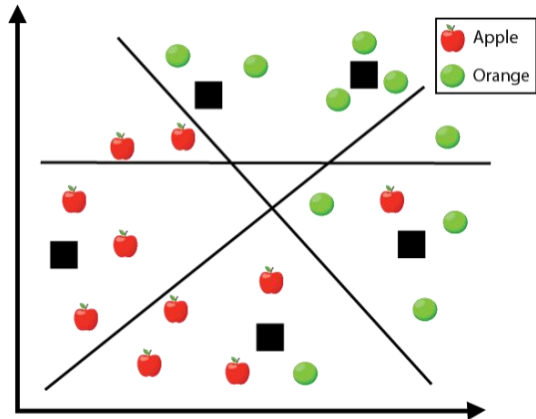
On the training data:
25% error



What is the “Learning” Problem?

Find the best line given
(only) the training data

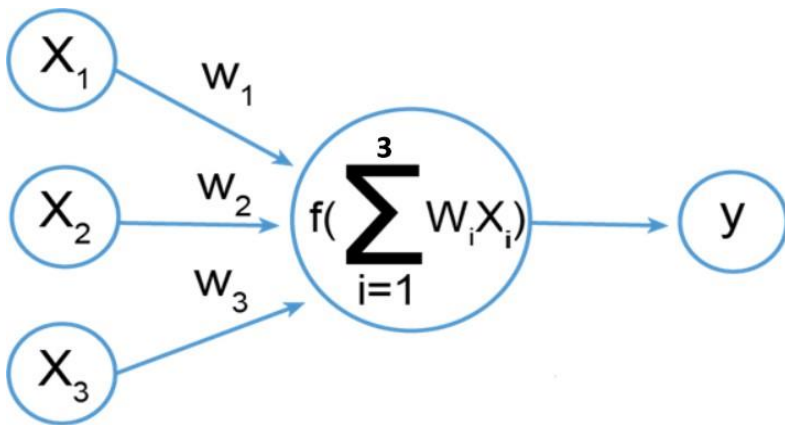
Find $\omega_1, \omega_2, \omega_0$ in
 $\omega_1 x_1 + \omega_2 x_2 + \omega_0 = 0$



Characteristics

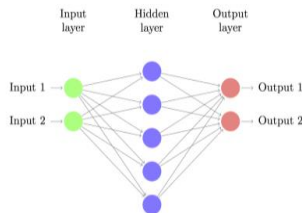
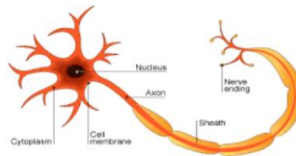
- ▶ **Linear Classifier:**
 - ▶ $(d-1)$ additions; d multiplications
- ▶ Simple at test time.
- ▶ Additional “offline” training to find w
 - N : Total number of samples
 - d : Total number of features or dimensionality

Understanding Linear Classifier as “Neuron”

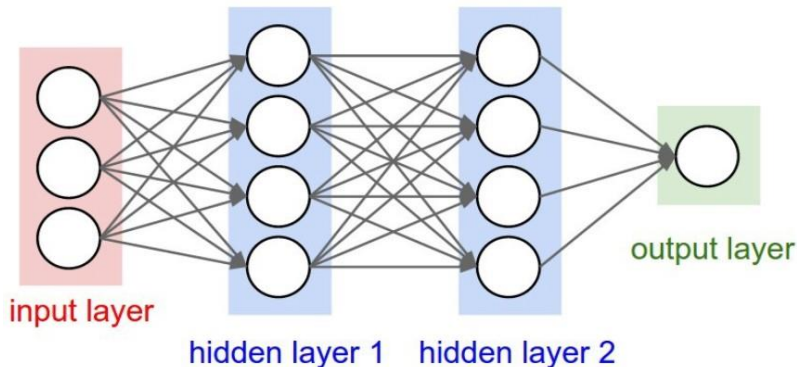


Neural Networks

- ▶ Biologically inspired networks.
- ▶ Complex function approximation through composition of functions.
- ▶ Can learn arbitrary Nonlinear decision boundary



A Peep into Deep Neural Networks



- ▶ Demo_LC_Penguins_data
 - ▶ Dataset : Penguins data
 - ▶ Objective : To apply Linear classifier to a multi-class classification problem

Thanks!!
Questions?